



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 4 Option Topics

9189/4

JUNE 2016 SESSION

1 hour 15 minutes

Additional materials:

Answer paper

Data booklet

Graph paper

Mathematical tables and/or electronic calculator

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper.

Answer a total of **four** questions. Do not answer more than **two** questions from any **one** Option.

You are advised not to attempt questions on Options for which you have not been prepared.

Write your answers on the separate answer paper provided.

Begin each answer on a fresh page.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

Where relevant, the symbols for aluminium, chlorine and iodine are printed as: Al; Cl; I; respectively.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 11 printed pages and 1 blank page.

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BIOCHEMISTRY

Not more than two questions to be answered from this Option.

- 1 (a) Phosphorylation of glucose is the first step of glycolysis and is catalysed by hexokinase and glucokinase.
- (i) Sketch a graph to show how the rate of glycolysis, catalysed by hexokinase, changes with the glucose concentration.
- (ii) Explain the shape of the graph.
- (iii) On the same graph, draw a line to show the effect of
1. a competitive inhibitor,
 2. a non-competitive inhibitor.
- [6]
- (b) Explain how a competitive inhibitor differs from a non-competitive inhibitor in their functions.
- [4]
[Total: 10]
- 2 (a) (i) Draw diagrams to show the differences between a phosphoglyceride and a triglyceride.
- (ii) Explain, with the aid of diagrams, how phosphoglycerides are important in the formation of micelles and bimolecular layers.
- [6]
- (b) (i) Using the Na^+/K^+ pump as an example, explain the concept of active transport.
- (ii) Name one other transport system used in living organisms.
- [4]
[Total: 10]

3 The major component of egg white is a globular protein, albumin. When heated, egg albumin undergoes irreversible change to a white protein.

- (a) (i) Define the term *globular protein*.
- (ii) Give the differences between a globular protein and a fibrous protein.
- (iii) Describe and explain the changes which occur when egg albumin is heated.
- (iv) Suggest how heavy metal ions would affect the structure and function of egg albumin.

[7]

- (b) The presence of albumin as one of the proteins in human blood plasma has been demonstrated by a process called electrophoresis.

Describe this process, stating clearly, the basis upon which separation of plasma components is achieved.

[3]

[Total: 10]

Not more than *two* questions to be answered from this Option.

4

(a) Define the terms

- (i) *sewage,*
- (ii) *biological oxygen demand, BOD.*

[2]

(b) Sewage is normally treated in three stages, namely, primary, secondary and tertiary treatments.

(i) Describe the processes involved in primary sewage treatment.

(ii) Fig. 4.1 shows a secondary sewage treatment plant.

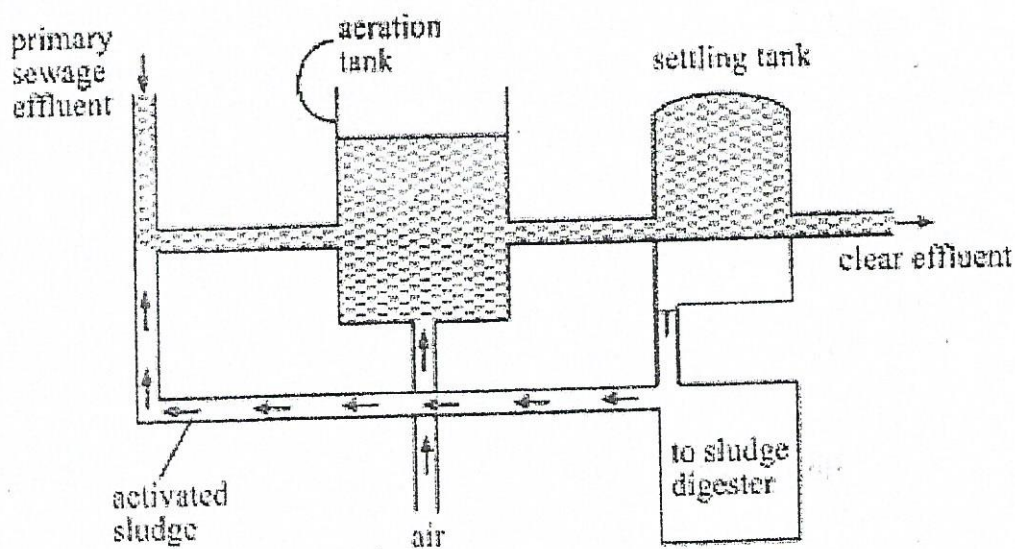


Fig. 4.1

1. State the main purpose of secondary sewage treatment.

2. Describe what happens in the aeration tank.

(iii) The clear effluent from the secondary treatment plant is transferred to the tertiary treatment plant in order to remove organic matter, phosphates and nitrates.

Describe how the effluent is treated to remove any **two** of these pollutants.

[7]

(c) State one side effect, in humans, of drinking chlorinated water.

[1]

[Total: 10]

- 5 (a) (i) Define the term *greenhouse gas*.
 (ii) Name any **two** greenhouse gases other than carbon dioxide.
 (iii) State a method that can be used to remove each of the greenhouse gases named in (ii). [5]
- (b) Give one human activity which is a source of greenhouse gases. [1]
- (c) Explain the greenhouse effect, using carbon dioxide as an example. [4]
 [Total: 10]
- 6 (a) Kaolinite is a typical 1:1 clay whilst montmorillonite is a typical 2:1 clay.
 Describe by means of diagrams the structures of the two clays. [6]
- (b) During heavy rainfall, leaching results in loss of essential ions.
 Explain why water soluble ions such as NH_4^+ or K^+ are not washed away in clay soils but could be lost in acidic rain. [4]
 [Total: 10]

Not more than two questions to be answered from this Option.

- 7 (a) The phase diagram for the lead-tin system is shown in Fig. 7.1.

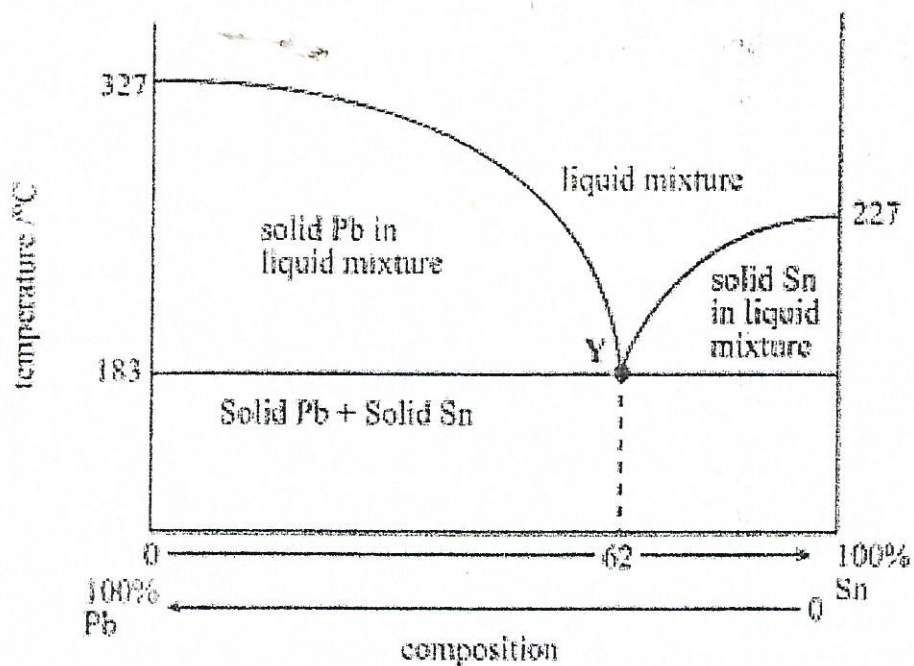


Fig. 7.1

- (i) State what point Y represents.
- (ii) Describe the composition of the first melt produced on heating a solid mixture of 20% tin and 80% lead.
- (iii) State, with reasons, the temperature at which the melt in a(ii) is formed.
- (iv) State and explain the effect of adding tin to lead on the freezing point of lead.

[6]

(b) The phase diagram for the copper-nickel system is shown in Fig. 7.2.

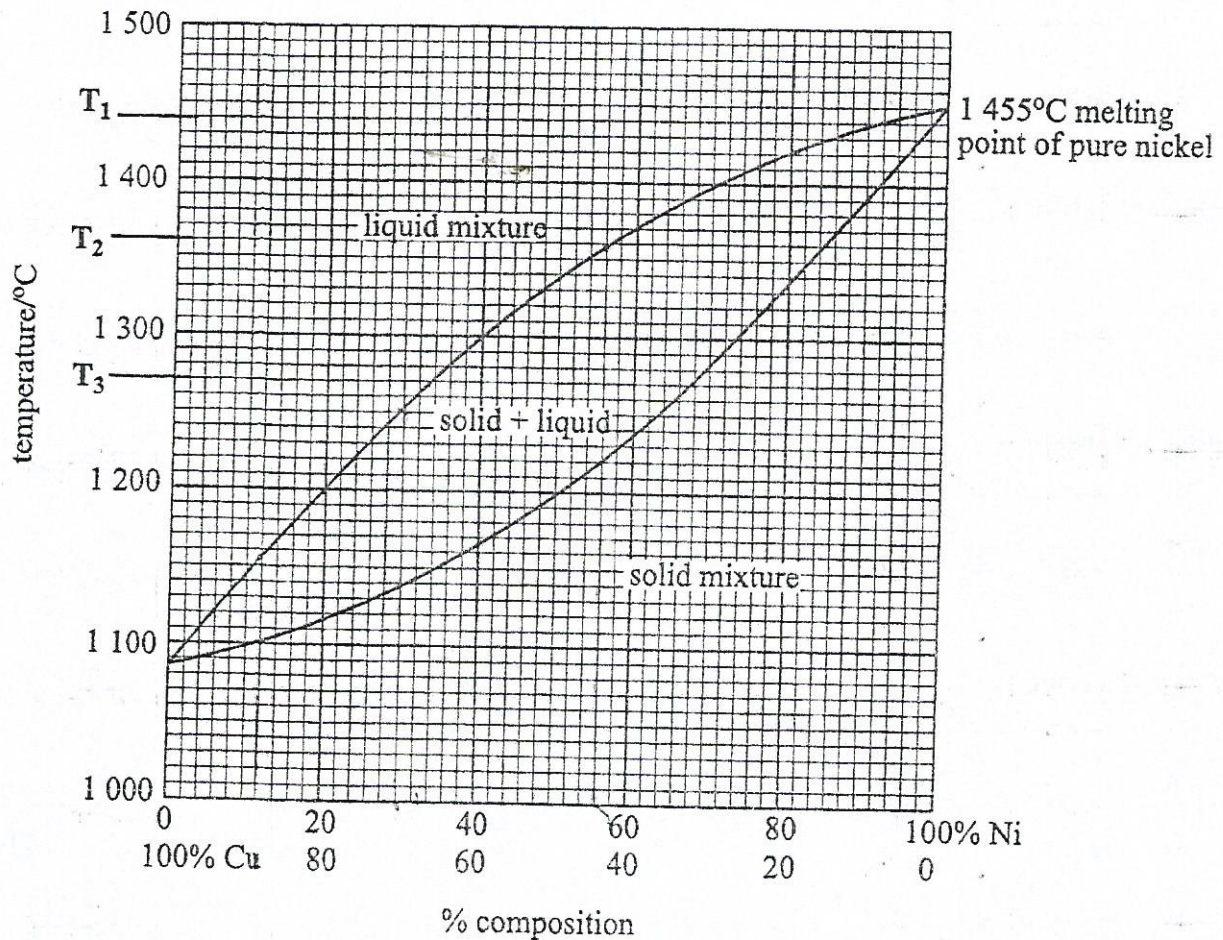


Fig. 7.2

A liquid mixture containing 30% nickel is cooled from temperature T_1 to T_3 .

- (i) State the temperature at which crystallisation begins.
- (ii) Give the composition of the solid present in the solid-liquid mixture at temperature

1. T_2 ,

2. T_3 .

[3]

- (c) Give any one difference between the Cu/Ni and Pb/Sn phase diagrams. [1]

[Total: 10]

8

(a) Define the term *partition*.

[1]

(b) (i) Describe a method which could be used to determine the partition coefficient for the distribution of iodine between ether and water.

(ii) The partition coefficient for the distribution of iodine between ether and water was found to be 42.7 at 25 °C. Titration of 25.00 cm³ of the ether layer required 26.50 cm³ of 0.0500 mol dm⁻³ sodium thiosulphate solution. 25.00 cm³ of the aqueous layer was titrated with 1.00 × 10⁻³ mol dm⁻³ sodium thiosulphate solution.

Calculate the

1. concentration of iodine in the ether layer.
2. concentration of iodine in the aqueous layer.
3. volume of sodium thiosulphate solution required for complete reaction with iodine in the aqueous layer.

[9]

[Total: 10]

- 9 (a) Fig. 9.1 shows how a continuous extraction of a non-volatile and heat stable solute is carried out.

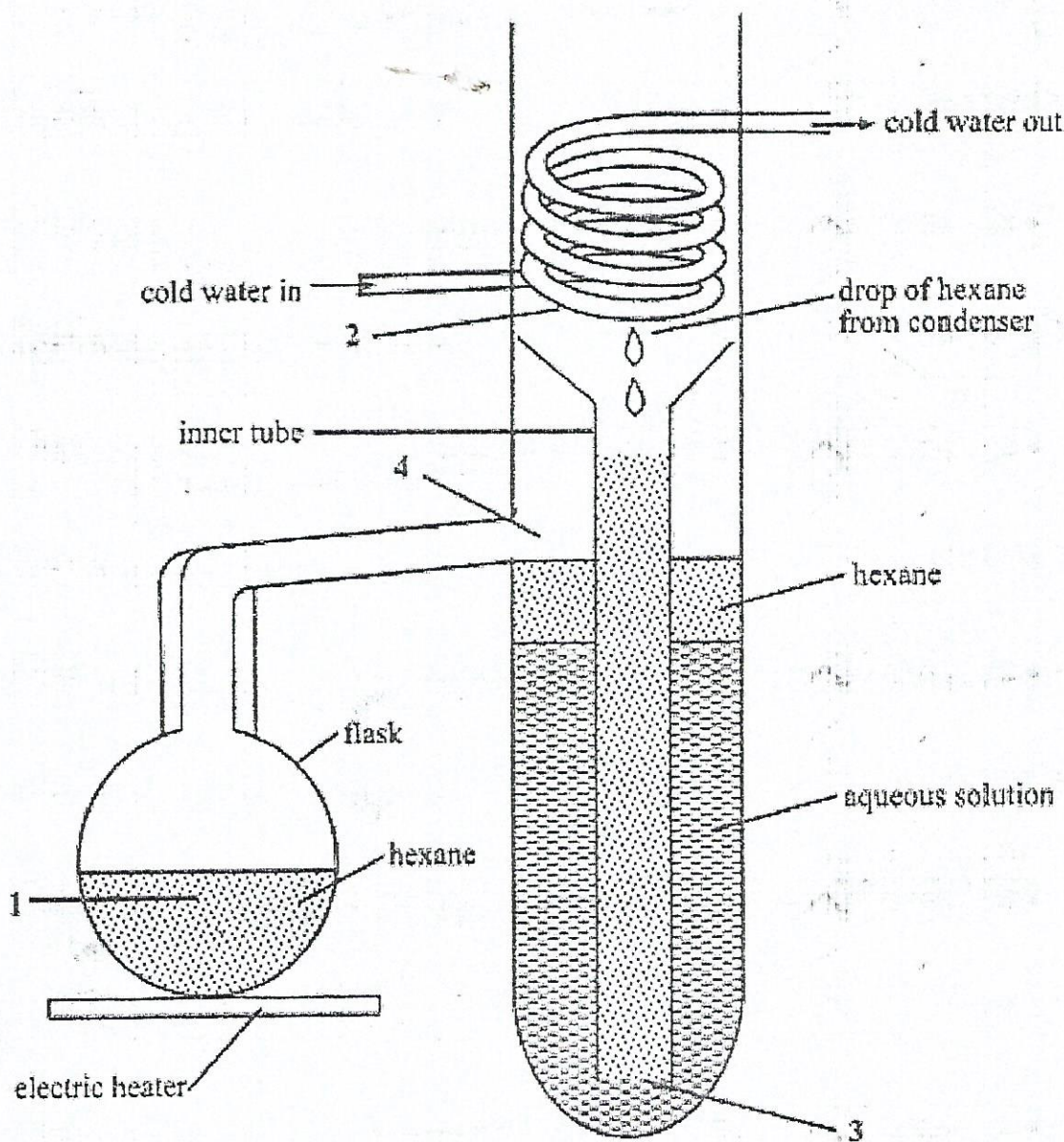


Fig. 9.1

At points 1, 2, 3 and 4 the processes taking place ensure continuity of separation.

Identify the process taking place at each of the four points and state how each process helps to extract the analyte. [6]

- (b) (i) Describe the principle of separation involved in steam distillation.
 (ii) Suggest how water is separated from the distillate of steam distillation.

[4]

[Total: 10]

TRANSITION ELEMENTS

Not more than two questions to be answered from this Option.

- 10 (a) (i) Define the term *disproportionation*.
(ii) Write an equation for the disproportionation of aqueous Cu (I) ions.
(iii) Calculate the E°_{cell} for the reaction in (ii).
(iv) Compare the relative stabilities of Cu (I) and Cu (II) ions in aqueous solution. [5]
- (b) Describe and explain, using equations, the observable changes that occur when excess $\text{KI}_{(\text{aq})}$ is added to a portion of $\text{CuSO}_{4(\text{aq})}$ followed by addition of $\text{Na}_2\text{S}_{2}\text{O}_{3(\text{aq})}$. [5]
- [Total: 10]
- 11 (a) Write the formula and state the colour of each of the following ions in aqueous solution.
(i) manganate (VI)
(ii) manganate (VII) [2]
- (b) Manganate (V) is another oxo-anion of manganese.
(i) Suggest the formula of manganate (V) ion.
(ii) Predict the shape of manganate (V), stating the bond angle. [3]
- (c) Manganate (VII) and manganese (IV) are products of disproportionation of manganese (VI) in acidic solution.
(i) Write a balanced chemical equation for the reaction.
(ii) Suggest how a solution of manganese (VI) may be prevented from undergoing disproportionation. [3]
- (d) Another method which is used to prepare manganate (VI) involves reacting manganate (VII) with a strong alkali such as potassium hydroxide.
Write a chemical equation for the reaction. [1]
- (e) Write an equation to show why solutions of $\text{Mn}^{3+}_{(\text{aq})}$ are highly acidic. [1]
- [Total: 10]

- 12 (a) Fig. 12.1 shows some of the reactions of cobalt (II) aqueous ions.

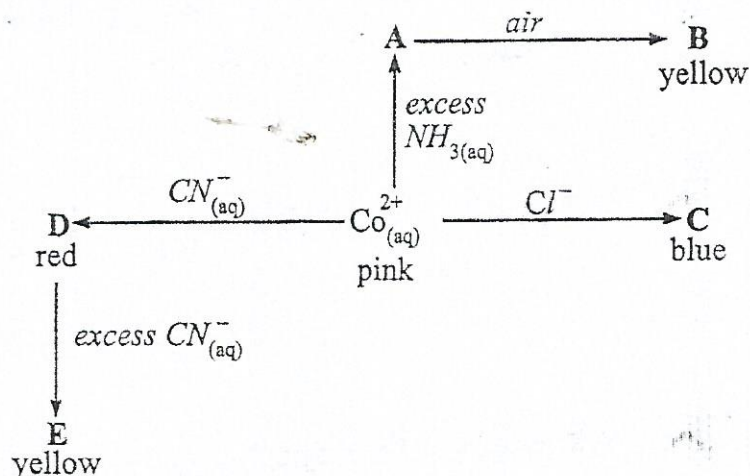


Fig. 12.1

- (i) Suggest the formulae of each of the complexes A, B, C, D and E.
- (ii) State two effects of replacing water ligands with chloride ligands on the $\text{Co}^{2+}_{(\text{aq})}$ ion other than change in colour.
- [7]
- (b) When air is bubbled through an aqueous solution containing CoCl_2 , NH_4Cl and NH_3 and the resulting solution evaporated, crystals of an octahedral complex ion, X, can be isolated. X has a charge of -1 and has the following composition by mass:

Co	25.2%
N	24.1%
H	5.1%
Cl	45.6%

- (i) Calculate the empirical formula of X.
- (ii) Name the ligands present in X.

[3]
[Total: 10]